MPDA: Multivariate Probability Distribution Autoencoder for Hyperspectral Anomaly Detection

Zhenhua Mu, Yihan Wang, Yating Zhang, Chuanming Song, and Xianghai Wang

***Abstract*— In recent years, the significant success of deep learning (DL) in computer vision has contributed to its continuous development in the field of hyperspectral image (HSI) anomaly detection (AD). However, in practical applications, HSI-AD based on DL faces many challenges due to the inability to effectively acquire training samples and predict the types of anomaly targets. This makes it a challenging task, especially for AD in complex scenes. In this article, we propose an unsupervised DL framework for HSI-AD based on the multivariate probability distribution autoencoder (MPDA). First, to explore the distribution characteristics of high-dimensional data, we use the probability density histogram to statistically distribute the frequencies of each interval adaptively, dividing the HSI and obtaining an AD-guided image through the designed grid structure. Second, we propose an unsupervised multilayer autoencoder network based on energy-weighted skip connections. By coupling the detection-guided image in the network, we achieve reverse-guided module reconstruction, weakening the feature representation of anomalous in the reconstructed information and enhancing the separability of targets. Finally, we model the reconstructed error images using the multivariate skewed t-distribution based on data distribution characteristics to obtain the final AD map. Through the comparative experiments with other innovative AD algorithms on authentic HSI datasets captured in five different scenarios, the proposed algorithm demonstrates strong generalization and detection capabilities. The source code of the MPDA will be public at https://github.com/muzhenhuam/MPDA/tree/master.**

***Index Terms*— Anomaly detection (AD), autoencoder, hyperspectral image (HSI), probability density distribution, skewed distribution, unsupervised deep learning (DL).**

I. Introduction

HYPERSPECTRAL image (HSI) is an important remote sensing data source with rich spectral information and spatial resolution [1], [2], widely used in fields such as land-cover classification, environmental monitoring, and resource management [3]. In remote sensing, the spectral reflectance features of different surface cover objects often exhibit significant variations [4]. These changes often contain relevant information about anomalous targets such as geological mineralization exploration, food quality pollution detection, and plant diseases [5], [6]. This anomaly information is different from the background elements of HSIs, which have the characteristics of a small proportion and a low probability of occurrence, cannot be identified before data processing, and so on. How to effectively detect the anomaly targets in HSIs under complex scenes is a challenging and crucial Earth observation task [7].

Probabilistic distribution-based anomaly detection (AD) methods form the foundation of traditional HSI-AD. The global RX (GRX) algorithm proposed by Reed and Yu [8], which assumes that the background information in HSIs follows the Gaussian distribution, determined the deviation of the test pixel from background samples covariance and mean to estimate the probability density function and ascertain the anomalous level of pixels. With the widespread demand for HSI-AD techniques in various fields, the GRX algorithm has been developed. The superpixel-based dual-window RX (SPDWRX) algorithm [9] partitions HSIs into fine superpixels based on the entropy method. By extending the minimum bounding rectangle to obtain background information for each superpixel, it performs local RX AD on each pixel within the same superpixel using the same background. Zhang et al. [10] proposed a fractional Fourier transform tensor RX (FrFT-TRX) algorithm to transform HSIs into the fractional Fourier domain. Subsequently, the tensor-based RX method is employed to detect the anomalous targets. The recursive RX with extended multiattribute profiles (RRXEMAPs) algorithm [11] combines extended multiattribute profiles with RX. It first extracts spatial structure information from HSIs using multiattribute profiles. Then, it utilizes an RX detector to remove pixels that may be anomalous targets, purifying background information. Finally, it uses the RX detector again between the extracted features of the multiattribute profile and the new background distribution to determine anomalies. RX-based algorithms usually assume data follows a multivariate normal distribution, sensitive to spectral variations. However, in reality, spectral variations of features on the ground are often complex and cannot be described by a single distribution model, leading to poor performance of the RX-based algorithms when dealing with nonuniform or non-Gaussian distributed data.

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Zhenhua Mu is with the School of Geography, Liaoning Normal University, Dalian 116029, China (e-mail: mudatoummm@gmail.com).

Yihan Wang and Yating Zhang are with the School of Computer Science and Artificial Intelligence, Liaoning Normal University, Dalian 116029, China (e-mail: yihanwang.lnnu@gmail.com; yt_zhang30@163.com).

Chuanming Song is with the School of Information Engineering, Dalian University, Dalian 116622, China (e-mail: chmsong@163.com).

Xianghai Wang is with the School of Geography and the School of Computer Science and Artificial Intelligence, Liaoning Normal University, Dalian 116029, China (e-mail: xhwang@lnnu.edu.cn).

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With the introduction and development of some new probability distribution functions, Mercan and Alam [12] modeled the background information of HSIs using stable distributions, considering anomalies in the distribution as regions of interest, which effectively detect anomalous points in HSIs by comparing regions of interest with known references using Euclidean distance. Mu et al. [13] obtained high- and low-frequency information from HSIs using the nonsubsampled Shearlet transform and then obtained the initial anomaly score image through the isolation forest model based on the spectral information divergence. Subsequently, different coefficient states were obtained through a mixture of Gaussian distributions. Finally, the probability matrix of model parameters was obtained through the expectation–maximization algorithm to obtain the anomaly result image. Yu et al. [14] proposed a probability distribution representation detector (PDRD), which used multivariate Gaussian distributions to represent HSIs, exploring spatial information of both background and anomalies. Kayabol et al. [15] used the multivariate skewed t-distribution (MVSkt) modeling. A variational Bayesian approach was used to estimate the distribution parameters and determine the distribution-based anomaly data. Li et al. [16] used a normalizing flow-based PDRD (NF-PDRD) for AD. The probability distribution representation of the mean vector and standard deviation vector of HSIs was first obtained using a variational autoencoder. The Gaussian distribution is subsequently normalized to a more complex distribution model using the Gaussian mixture model estimation strategy to extract the probability representation of complex distribution models to complete the AD.

In recent years, with the development of deep learning (DL), HSI-AD methods based on deep generative models have attracted attention. Gao et al. [17] proposed a self supervised learning network framework, which utilizes self-supervised training networks to generate training patches with spots from single-band HSI, reconstructing each pixel using its neighboring pixels. The reconstruction error effectively separates anomalous target points from background information. Li et al. [18] combined low-rank representation with DL to efficiently solve the low-rank representation model using an alternating multiplicative optimizer and incorporated the solution results as prior knowledge into the deep network to guide parameter optimization and obtain anomaly results. Duan et al. [19] proposed an HSI-AD method based on collaborative representation neural networks (CRNNs). First, the global and local dictionaries of HSIs were constructed, and further, they used a dual-stream network to learn the global and local features of HSIs. The final detection result image was obtained by fusing the residual images of the original data and the dual-stream data. Lin et al. [20] introduced a dynamic low-rank and sparse priors-constrained nonlinear-based deep autoencoder (DLRSPs-DAE) model. This model designed two types of autoencoders: one is a deep autoencoder based on low-rank priors to suppress anomalous information reconstructing background targets and the other is a sparse prior constrained deep autoencoder obtained from HSI decomposition, which reflects the sparsity of anomalous targets in HSIs. A nonlinear fusion strategy is used to fuse the two encoder results to obtain an AD image. Wang et al. [21] proposed a fully convolutional autoencoder network based on an attention gate connection. First, the encoder generated low-dimensional features. Then, in the decoder stage, the attention gate features and upsampled deep features were combined through skip connections to reconstruct background information. Finally, the guided filtering was applied to the reconstruction error to obtain anomalous pixels. Fan et al. [22] proposed a robust graph autoencoder (RGAE) framework for AD. First, a robust autoencoder framework is proposed. Then, a graph regularization term based on hyperpixel segmentation is embedded in the autoencoder. Finally, the framework is used for AD. Xiang et al. [23] proposed a guided autoencoder-based detector (GAED) for HSI-AD. First, a reconstruction HSI is obtained using a skip-connected multilayer autoencoder. Second, a guided image is incorporated into the network to reduce the feature representation of the anomalous target by providing feedback information. Finally, we use the reconstruction error data for AD. Wang et al. [24] proposed a pixel-shuffle downsampling spot reconstruction network, which utilized a receptive field spotstructure to reconstruct the spectral information of central pixels using neighboring pixels, leading to higher reconstruction errors for anomalous pixels. Furthermore, the pixel-shuffle downsampling method was introduced to break the spatial correlations between anomalous target pixels, highlighting the anomalous targets. In HSIs, anomalous pixels are often much less than background pixels, leading to a severe sample imbalance problem in the dataset. How to suppress background and highlight anomalous targets in complex application scenarios still has many challenges and merits further research.

This article proposes an HSI-AD method based on a multivariate probability distribution autoencoder (MPDA). First, to explore the distribution characteristics of high-dimensional data, the probability density distribution histogram is used to statistically count the frequency of each interval, adaptively partitioning the HSI into rows and columns to form the grid structure. The grid structure mapping from an unordered to an ordered form of HSIs leads to intuitive differences between background information and anomalous targets, thereby obtaining an AD-guided image. Second, a multilayer autoencoder network based on energy-weighted skip connections is proposed. By coupling the autoencoder system with the reverse guided module, the autoencoder network will give more weight to the background when reconstructing the background information, reducing the anomalous features and increasing the separability of targets. Finally, the reconstructed error image is modeled using a multivariate skewed t-distribution according to data distribution characteristics to further improve the accuracy of AD and obtain the final AD image. This method effectively highlights anomalous target situations while suppressing background information, enhancing the accuracy of AD. The main contributions of this article are summarized as follows.

1) A grid structure is designed to divide the HSI into rows and columns, achieving a mapping from an unordered to an ordered form of HSIs. The differences in quantity, density, spectral trends, and forms of each grid

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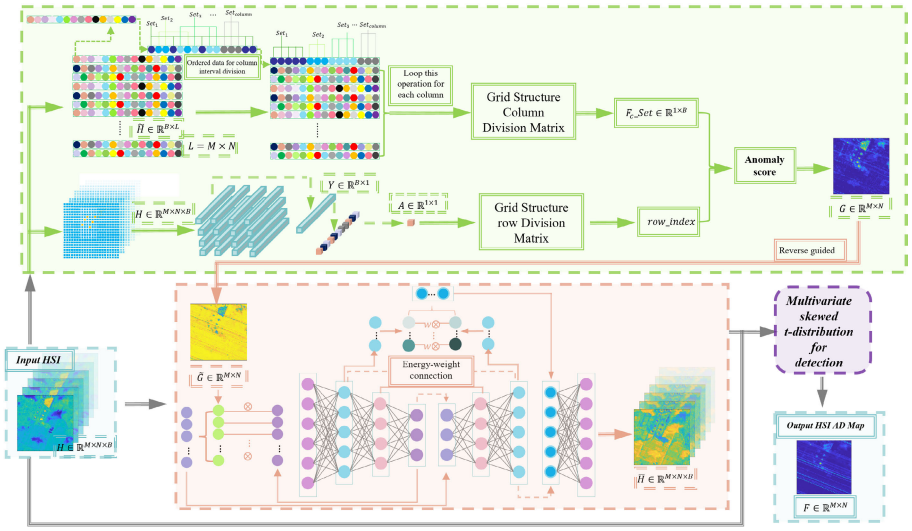


Fig. 1. Flowchart of HSI-AD-MPDA framework in this article.

region lay the foundation for the effective separation of background and anomaly. By quantifying the deviation values between test points and background intervals, the differences in quantity and density between anomalies and background are effectively calculated to obtain an AD-guided image. In addition, this grid structure can also overcome the inherent spectral redundancy information in HSIs.

2) A multilayer autoencoder network based on energy-weighted skip connections is proposed for HSI-AD without any prior information constraints, effectively learning the intrinsic features of the data without the need for explicit data labels.
3) The HSI-AD-MPAD method is proposed. By combining the autoencoder system with a multivariate skewed t-distribution model, the hidden spatial intrinsic features are discovered and the spatial information is better utilized. This method has strong generalization and detection capabilities, making it suitable for AD in complex scenarios.

The remainder of the article is organized as follows. Section II presents the details of the MPDA framework. Section III presents the experimental results and performance analysis. Section IV presents the conclusion of the algorithm.

II. Proposed Methodology

This article proposes the HSI-AD framework based on the MPDA, which consists of three parts as shown in Fig. 1. First, by utilizing the probability density histogram to statistically count the frequency of each interval, describing categorical data, and exploring the distribution characteristics of HSIs, the data is adaptively partitioned into a grid structure to obtain the guided image for AD. Then, through unsupervised DL reconstruction of the multilayer autoencoder based on energy-weighted skip connections is performed, and coupling the autoencoder with the reverse guided module to obtain the reconstructed background information. Finally, the error data is modeled using a multivariate skewed t-distribution to obtain the final AD image. Section II-A–II-E, respectively, analyze and explain the three parts of the framework.

A. Guided Image Extraction for HSI-AD Based on Grid Structure Column Division

It is assumed that the given HSI band group is denoted as $H \in \mathbb{R}^{M\times N\times B}$, where $M \times N$ is the size of the band image and B is the number of bands in the band group. Adaptive partitioning of the data using grid structure, first, H is transformed into $\widetilde{H} \in \mathbb{R}^{B\times L}(L = M \times N)$. Second, the unordered data points of $\widetilde{H}_b \in \mathbb{R}^{1\times L}(b \in \{1, 2, \ldots, B\})$ are taken to be ordered, and the ordered point set is divided into columns of the grid structure according to num_{col} intervals. The number of pixel points in each interval is called frequency, and the mean value of pixel points in each interval is called center position. The frequency number and center position of each interval are statistically distributed using the probability density histogram, and the statistics are looped B times, and finally the column frequency matrix $F_c \in \mathbb{R}^{B\times \text{num}_{\text{col}}}$ and the interval column center position matrix $\text{ICP}_c \in \mathbb{R}^{B\times \text{num}_{\text{col}}}$ are obtained after the column division, where num_{col} denotes the number of columns in the grid distribution, which is calculated by (1), and p is the probability empirical value is set to 0.9. Fig. 2 represents the process of the guided map column division.

First, it is necessary to decide to divide the data into num_{col} intervals; then determine the width of the interval of the column, interval width = data range/number of intervals, the data range is the difference between the maximum and minimum values of the data; and then, starting from the minimum value of the data, divided into intervals of equal spacing; and finally, calculate the column frequency matrix F_c and the interval column center position matrix ICP_c. For each interval, the number of data points that fall into that interval is counted, and this number is the frequency of that interval. The center position of the interval is the midpoint of each interval; center position = (lower limit of the interval + upper limit of the interval)/2.

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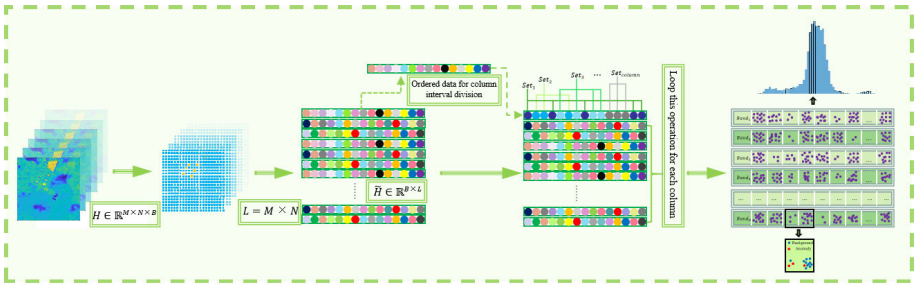


Fig. 2. Schematic of HSI-AD-guided image column division.

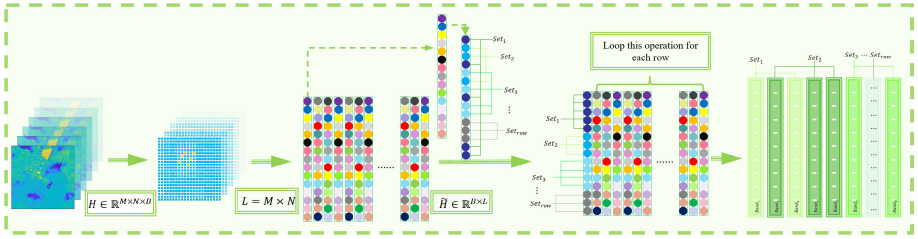


Fig. 3. Schematic of HSI-AD-guided image row division.

According to the characteristics of anomalous targets with low probability of occurrence and small area. We believe that the frequency corresponding to each grid reflects the quantity and density of data points, and the smaller the frequency of the grid of the $\widetilde{H}_b$ where a data point is located, the higher the probability that it is an anomalous target. It is worth noting that for the column division of the grid, due to the lack of prior information, it cannot be guaranteed that anomalous points and background information points are completely separated within each interval. In some cases, as shown in Fig. 2, both types of data points may be mixed in the same grid. To more accurately separate background and anomalous target points and enhance the accuracy of the guided map, we can reduce the separation error between anomalous points and background information by adjusting the row division of the grid, thereby enhancing the separability of the data and obtaining more reliable detection results

$$\text{num}_{\text{col}} = p \times M \times N. \tag{1}$$

B. Guided Image Extraction for HSI-AD Based on Grid Structure Row Division

The unordered data points of $\widetilde{H}_l \in \mathbb{R}^{B\times 1}$ are arranged in order, where $l \in \{1, 2, \ldots, L\}$, and the ordered point set is divided into rows of the grid structure according to num_{row} intervals. Fig. 3 represents the process of row division of the HSI-AD-guided map. Similar to column division, row division divides the data into num_{row} intervals, $\widetilde{H}_l$ uses the probability density histogram to statistically distribute the frequency and center position of each interval to obtain the row frequency matrix $F_r \in \mathbb{R}^{1\times \text{num}_{\text{row}}}$ and the interval row center position matrix $\text{ICP}_r \in \mathbb{R}^{1\times \text{num}_{\text{row}}}$ after row division. Let

$$\begin{cases} \text{num}_{\text{row}} = \text{num}(CC \geq q) \\ CC = \dfrac{\sum\left(H_{\tilde{b}} - \overline{H}_{\tilde{b}}\right)\left(H_{\tilde{b}+1} - \overline{H}_{\tilde{b}+1}\right)}{\sqrt{\sum\left(H_{\tilde{b}} - \overline{H}_{\tilde{b}}\right)^2 \sum\left(H_{\tilde{b}+1} - \overline{H}_{\tilde{b}+1}\right)^2}} \end{cases} \tag{2}$$

where num_{row} denotes the number of rows in the grid distribution, CC is the value of the correlation coefficient between the two bands $H_{\tilde{b}}$ and $H_{\tilde{b}+1}(\tilde{b} \in \{1, 2, \ldots, B-1\})$, $\overline{H}_{\tilde{b}}$ is the pixel mean value of the $H_{\tilde{b}}$ band, $\overline{H}_{\tilde{b}+1}$ is the pixel mean value of the $H_{\tilde{b}+1}$ band, and q is the probabilistic empirical value which we set to 0.98.

C. Guided Image Calculation

Assuming that $Y_l \in \mathbb{R}^{B\times 1}$ represents the H spectral dimension information at different positions, the anomaly score value anomaly_score of Y_l is calculated by dividing the grid into rows and columns to obtain the guided image $G \in \mathbb{R}^{M\times N}$. Let $Y_l = \left[y_1, y_2, \ldots, y_B\right]$, where $y_b \in \mathbb{R}^{1\times 1}$ is the test pixel point. Fig. 4 shows the flowchart of HSI-AD-guided map calculation based on grid division. The grid structure division achieves the mapping of HSIs from an unordered to an ordered form. Due to differences in quantity, density, spectral trends, and forms among each grid region, background information and anomaly targets exhibit intuitive distinctions, facilitating the separation of anomalous data points and enhancing discriminability. By tracking y_b in the grid division of the ICP_c to find the corresponding center position matrix $y_\text{ICP} \in \mathbb{R}^{1\times \text{num}_{\text{col}}}$ through the (3) calculation to get the minimum numerical error of the column index position c_index

$$[\sim, c_\text{index}] = \min|y_\text{ICP}(:) - y_b|. \tag{3}$$

The column frequency value $y_b_F_c \in \mathbb{R}^{1\times 1}$ corresponding to F_c is given based on the grid position c_index, where y_b is located in each band. Construct a column frequency set F_c_Set under band by band detection: $F_c_\text{Set} = \left[y_1_F_c, y_2_F_c, \ldots, y_B_F_c\right]$, $F_c_\text{Set} \in \mathbb{R}^{1\times B}$.

Iterate through the grid row structure from 1 to num_{row} to determine the row index. $n(n \in \{1, 2, \cdots, \text{num}_{\text{row}}\})$ times in the loop, corresponding $n_F_r \in \mathbb{R}^{1\times 1}$ and $n_\text{ICP}_r \in \mathbb{R}^{1\times 1}$ are found in F_r and ICP_r based on the position tracking n.

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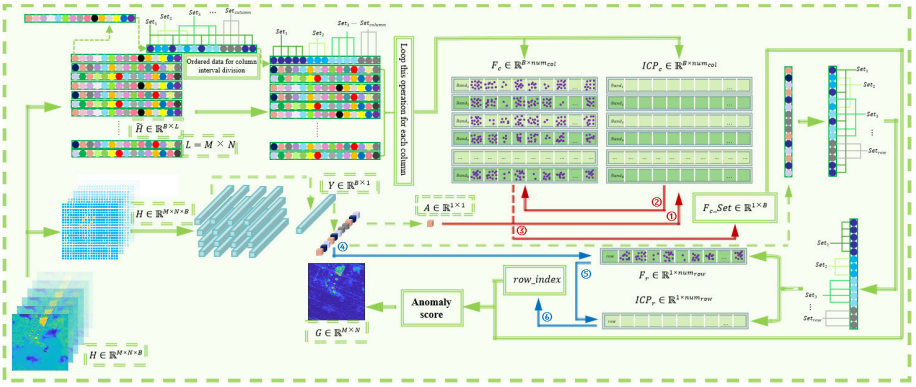


Fig. 4. Flowchart for the computation of the HSI-AD-guided image based on grid segmentation.

The index matrix row_index $\in \mathbb{R}^{B\times 1}$ of the row positions of the elements in the original matrix after ascending ordering of the numerical errors is obtained by the calculation of (4), and sort$(\cdot)$ is the function of ascending ordering of the elements in a given interval

$$[\sim, \text{row_index}] = \text{sort}(|Y_l - n_\text{ICP}_r|). \tag{4}$$

The optimal separable band index position band_index is determined based on row_index and n_F_r. The acquisition of the optimal separable band can effectively highlight anomalous target points and increase the separability between background and anomalies. Equation (5) calculates the minimum column frequency min_F_c corresponding to the separable band and the corresponding index position min_p

$$[\text{min}_F_c, \text{min}_p] = \text{min}(F_c_\text{Set}(\text{band_index})). \tag{5}$$

The minimum separated band position min_band is found by min_p and obtains the maximum frequency value max_F_c in the corresponding F_c for that band, and the anomaly ratio value score of the pixel is calculated according to the following equation:

$$\text{score} = \text{max}_F_c/\text{min}_F_c. \tag{6}$$

Calculate the ratio between the maximum frequency number max_F_c and the minimum frequency number min_F_c of the test pixel, quantifying the deviation value between that point and the high-density, high-quantity background interval. Effectively calculate the difference between anomaly and background in terms of quantity and density. This process not only facilitates the accurate separation of anomaly target and background information but also overcomes the inherent spectral information redundancy characteristic of HSIs and effectively highlights anomaly information for the detection task. Traverse each row to compute score and sum cumulatively to obtain anomaly_score for different positions of Y_l. After the traversal is completed, the guided image G is obtained.

Probability density histograms can help identify abnormal information in HSIs to a certain extent. By looking at the histogram, it is possible to identify the main peaks (normal data) and the tails (possible anomalies) of the data. The probability density histogram can be used as a preliminary tool for HSI-AD to help identify some obvious anomalous information. However, due to the high-dimensional and complexity of HSIs, relying on the histogram alone to detect anomalies has greater limitations, so we will only obtain images through the grid structure of the probability density histogram as the guided image, laying the foundation for the following branches to better highlighting of anomalous information.

The extraction method of the guided image is as summarized in Algorithm 1.

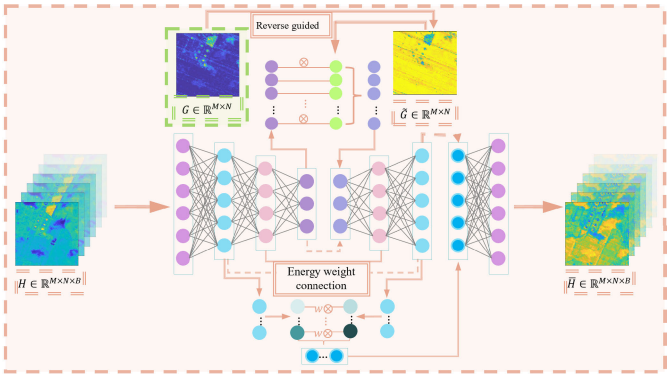


Fig. 5. Illustration of multilayer autoencoder reconstruction based on energy-weight skip connections.

D. Multilayer Autoencoder Reconstruction Based on Energy-Weighted Skip Connections

An autoencoder [25] is usually used as an unsupervised DL method to mine the intrinsic features of data in the absence of data labels. In this section, we propose a multilayer autoencoder background reconstruction unsupervised learning method based on energy-weighted skip connections, as shown in Fig. 5.

It is assumed that $H \in \mathbb{R}^{M\times N\times B}$ are transformed to $\grave{H} = [h_1, h_2, \ldots, h_T] \in \mathbb{R}^{B\times T}$, where T is the total number of pixels of HSIs. For the tth pixel $h_t (t \in \{1, 2, \ldots, T\})$, the output of each layer after the fully connected layer and activation function is shown in the following equation:

$$\begin{cases} z_t^u = W^u o_t^{u-1} + a^u \\ o_t^u = 1/\left(1 + e^{-z_t^u}\right) \end{cases} \tag{7}$$

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Algorithm 1 Guided Image Calculation

Input: $H \in \mathbb{R}^{M\times N\times B}$ - HSI with the size of $M \times N \times B$
Output: Guided Image $G \in \mathbb{R}^{M\times N}$ - Guided image with the size of $M \times N$

Initialize: p=0.9, q=0.98;
3D data H are transformed to 2D data $\widetilde{H} \in \mathbb{R}^{B\times L}$, num_{col} is calculated by Eq. (1), num_{row} is calculated by Eq. (2);
for $i \leftarrow 1$ **to** B **do**
  $X = \widetilde{H}(i,:)$, $X \in \mathbb{R}^{1\times L}$;
  X use probability density histograms to statistically distribute the column frequency matrix $F_c(i) \in \mathbb{R}^{1\times num_{col}}$ and column center position matrix $ICP_c(i) \in \mathbb{R}^{1\times num_{col}}$ of each band;
end for
Obtain $F_c \in \mathbb{R}^{B\times num_{col}}$ and $ICP_c \in \mathbb{R}^{B\times num_{col}}$;
for $i \leftarrow 1$ **to** L **do**
  $Y = \widetilde{H}(:,i)$, $Y \in \mathbb{R}^{B\times 1}$;
  Y use the statistical distribution of the probability density histogram to obtain the row frequency matrix $F_r \in \mathbb{R}^{1\times num_{row}}$ and the row center position matrix $ICP_r \in \mathbb{R}^{1\times num_{row}}$;
  $anomaly_score$=0, F_c_Set=[ ];
  for $j \leftarrow 1$ **to** B **do**
    $y = Y(j)$, $y \in \mathbb{R}^{1\times 1}$;
    $y_ICP = ICP_c(j,:)$;
    Calculate the column index value c_index by Eq. (3);
    Find the frequency value $y_j_F_c$ corresponding to the column frequency matrix by (j, c_index) position, $y_j_F_c$=$F_c(j, c_index)$;
    Store the frequency value $y_j_F_c$ in the F_c_Set, F_c_Set=[F_c_Set, $y_j_F_c$], $F_c_Set \in \mathbb{R}^{1\times B}$;
    for $n \leftarrow 1$ **to** num_{row} **do**
      n_F_r=$F_r(n)$;
      n_ICP_r=$ICP_r(n)$;
      Calculate the row index matrix row_index by Eq. (4);
      Calculate $score$ through Eq. (6);
    end for
    $anomaly_score$=$anomaly_score$+$score$;
  end for
  $result(i)$=$anomaly_score$, $result(i) \in \mathbb{R}^{1\times L}$;
end for
$result$ transformed to 2D data $G \in \mathbb{R}^{M\times N}$;

where o_t^u is the output data of each layer, W^u and a^u correspond to the weight and bias of each layer in the autoencoder, respectively, and $u \in \{1, 2, \ldots, 7\}$ denotes the number of layers. It is worth noting that the input $o_t^1 = h_t$ for the multilayer autoencoder network is based on energy-weighted skip connections.

The guided image computed in Section II-C is used in the fourth layer in the middle of the network in this section for the reverse guided module, and the embedding process is shown in the following equation:

$$\tilde{o}_{\mathrm{t}}^4 = \beta \times \tilde{g}_{(\mathrm{ii,jj})} \times o_{\mathrm{t}}^4 \tag{8}$$

where β is the penalty coefficient, in the reverse-guided module, the guided image G was subjected to an inverse phase subtraction operation by subtracting the pixel value of the guided image from the value 1 to obtain the inverse guided image $\widetilde{G}$, $\widetilde{g}_{(ii,j\ j)}$ is the pixel value at position (ii, jj) in $\widetilde{G}$, and $\tilde{o}_{\mathrm{t}}^4$ is the hidden layer after reverse guided module. The value of the horizontal coordinate ii ranges from 1 to the length of the network layer S, and the value of the vertical coordinate jj ranges from 1 to the width of the autoencoder layer D. The autoencoder network will give more weight to the background when reconstructing the background information based on the reverse guided module, thus reducing the anomalous targets, and it can reconstruct the background data in a better way.

The output representation of the fifth layer in a multilayer autoencoder network based on energy weight skip connections is shown in the following equation:

$$\begin{cases} z_t^6 = W^6\left(w_e_t^5 \times o_t^5 + w_e_t^3 \times o_t^3\right) + a^6 \\ o_t^6 = 1/\left(1 + e^{-z_t^6}\right) \end{cases} \tag{9}$$

where $w_e_t^5$ and $w_e_t^3$ are the energy value weights based on o_t^5 and o_t^3, respectively, and the calculation process is shown in (10). E_t^5 and E_t^3 are the energy values based on o_t^5 and o_t^3, respectively. The energy value calculates the area energy of the pixel, and the higher the energy represents the higher possibility that the changed pixel is a background target and vice versa. The calculation based on energy weights can further improve the accuracy of AD and lay the foundation for reconstructing the background information data

$$\begin{cases} E_t^5 = \left|\sum_{ii=1}^{S}\sum_{jj=1}^{D} -o_t^5(ii, j\ j) \times \log o_t^5(ii, j\ j)\right| \\ E_t^3 = \left|\sum_{ii=1}^{S}\sum_{jj=1}^{D} -o_t^3(ii, j\ j) \times \log o_t^3(ii, j\ j)\right| \\ w_e_t^5 = E_t^5/\left(E_t^5 + E_t^3\right) \\ w_e_t^3 = 1 - w_e_t^5. \end{cases} \tag{10}$$

The output of the sixth layer is shown in (11), and the computation of $w_e_t^6$ and $w_e_t^2$ is modeled in (10). o_t^7 is the output of the sixth layer network, which is also the reconstructed information data. The 2-D data o_t^7 is transformed into 3-D reconstructed background information $\bar{H} \in \mathbb{R}^{M\times N\times B}$

$$\begin{cases} z_t^7 = W^7\left(w_e_t^6 \times o_t^6 + w_e_t^2 \times o_t^2\right) + a^7 \\ o_t^7 = 1/\left(1 + e^{-Z_t^7}\right). \end{cases} \tag{11}$$

E. Multivariate Skewed t-Distribution Model HSI-AD Estimation

Probabilistic statistical distributions are usually robust to outliers in hyperspectral AD data. For this reason, we use different statistical distributions to detect anomalous data directly, as shown in Table I, among the five groups of data, except

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TABLE I
PERFORMANCE COMPARISON OF STATISTICAL DISTRIBUTIONS

Datasets	$AUC_{(PD,PF)}$					
	LA Airport	Pavia Center	LA Urban	MUUFL Gulfport	Spec TIR	Mean
Cauchy	**0.9160**	0.9331	0.9431	0.6083	0.9269	0.8654
Jeffreys	0.9159	0.9331	0.9431	0.6084	0.9270	0.8655
Laplace	0.9150	0.9415	0.9430	0.6194	0.9370	0.8711
Student t	**0.9160**	0.9331	0.9431	0.6083	0.9269	0.8654
Skewed t	0.9149	**0.9809**	**0.9434**	**0.6168**	**0.9443**	**0.8801**

for the LA Urban data, the other four groups in the skewed t-distribution have the best performance of AD, and the mean $AUC_{(PD,PF)}$ [26] is also the best. Although the Cauchy distribution has a heavy-tailed property, its skewness property is not as flexible as the skewed t-distribution. Moreover, the Cauchy distribution is very robust to outliers, but its heavy-tailed property may lead to a less stable model. Jeffreys distribution is mainly used as a prior distribution in Bayesian inference, which is less adaptive to the skewed properties of the data. Robustness is mainly in the Bayesian framework rather than dealing with outliers in the data. The Laplace distribution is more robust to outliers but is not suited to deal with biased properties in the data. It is robust to outliers but is not flexible enough to deal with complex patterns of anomalies in high-dimensional data. Student t-distribution has better robustness to outliers, but it is less capable of dealing with bias in the data and is not flexible to deal with it when compared to the skewed t-distribution. The skewed t-distribution has greater flexibility and can be adapted to the skewed characteristics of the data by adjusting the skewness parameter. This makes it more advantageous in dealing with asymmetric anomalies that are common in hyperspectral data. It is robust enough to handle outliers and anomalies in the data, and at the same time, due to its high flexibility, it can adapt to changes in the data distribution.

In summary, the skewed t-distribution has more flexibility and adaptability in HSI-AD and is better able to deal with skewed properties and complex anomaly patterns in the data. This makes it potentially more effective than other distribution models in practical applications.

For the source HSI spectral band group H, the reconstructed background data for a multilayer autoencoder based on energy-weighted skip connections as $\bar{H}$. Calculate the error data $K \in \mathbb{R}^{M\times N\times B}$, $K = H\text{-}\bar{H}$, $k_i \in \mathbb{R}^B$ is the ith pixel of K, where $i \in \{1, 2, \ldots, V\}$ is the pixel index by dictionary. Then, the differential data k_i is filtered using the 3×3 local neighborhood standard deviation. If the local standard deviation of the filter is greater than that of the background, the pixel is considered an anomaly target point. The filtered anomaly-highlighted error data is denoted as $a_k_i \in \mathbb{R}_+^B$. Considering the flexibility of parameter settings and robustness of the MVSkt, we model a_k_i with MVSkt [27] to obtain the following equation:

$$a_{k_i} = m + z \cdot b_\mathrm{v} + \sqrt{z} \cdot TT^{-1/2} \cdot \omega_i \tag{12}$$

where $\omega_i \sim \mathcal{N}(0, I)$ is a Gaussian random vector, a generalized inverse Gaussian random variable is defined as $z \sim \mathrm{GJG}(\alpha, \delta, \gamma)$, and b_v is a skewness vector. Define a normal-Wishart joint prior, m is the mean vector and TT is the precision (inverse covariance) matrix as $\{m, TT\} \sim \mathcal{N}\left(m_0, (\lambda TT)^{-1}\right)\mathcal{W}^{-1}\left(((v_0 - B - 1)\psi_0)^{-1}, v_0\right)$, where m_0, λ, v_0, and ψ_0 are hyperparameters. By integrating (12) over the latent variable z, we obtain the hyperbolic distribution for a_{k_i} [28] as follows:

$$f\left(a_{k_i} \mid m, TT; \alpha, \delta, \gamma\right) = \frac{\gamma^{\frac{\alpha}{2}}|TT|^{\frac{1}{2}}\exp\left\{\left(a_{k_i}-m\right)^T TT b_v\right\}}{\delta^{\frac{\alpha}{2}}K_\alpha(\sqrt{\delta\gamma})(2\pi)^{\frac{B}{2}}}\left(\frac{\delta + A_{k_i}}{\gamma + b_v^T TT b_v}\right)^{\frac{\alpha}{2}-\frac{B}{4}} K_{\alpha-\frac{1}{2}}\left(\sqrt{\left(\delta + A_{k_i}\right)\left(\gamma + b_v^T TT b_v\right)}\right) \tag{13}$$

where $A_{k_i} = \left(a_{k_i} - m\right)^T TT\left(a_{k_i} - m\right)$ and $K_\alpha(\cdot)$ is the second kind of modified Bessel function. If, in the hyperbolic distribution defined in (13), γ reaches infinity at $\alpha < 0$, we obtain the MVSkt

$$f\left(a_{k_i}\right) = \frac{|TT|^{\frac{1}{2}}\Gamma(-\alpha + B/2)\exp\left\{\left(a_{k_i} - m\right)^T TT b_v\right\}}{\delta^\alpha\Gamma(-\alpha)(\pi)^{\frac{B}{2}}} \times \left(\delta + A_{k_i}\right)^{\alpha-\frac{B}{2}}. \tag{14}$$

The parameter $\left\{m, TT, z, a_{k_{1:V}}\right\}$ is obtained by the variational Bayesian method [29], and the specific estimation steps are detailed in [27]. After parameter estimation, by taking the negative logarithm of the probability density function in (14), the AD score based on the MVSkt can be determined by the following equation, and the final AD image $F \in \mathbb{R}^{M\times N}$ is obtained by transformation:

$$\mathrm{AD}\left(a_{k_i}\right) = -\frac{2\alpha + B}{2}\log\left(1 + \frac{A_{k_i}}{\delta}\right) - \left(a_{k_i} - m\right)^T TT\mathrm{b_}v. \tag{15}$$

Based on the above analyses, this article proposes an HSI-AD-MPDA and the MPDA method can be summarized as Algorithm 2.

III. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

Section III reviews the generalization and detection accuracy of the MPDA by comparing its performance with different HSI-AD algorithms on five real datasets [30], [31]. First, the experimental datasets are introduced. Second, the parameters of the proposed algorithm are determined, and comparative experiments are analyzed. Then, the subjective and objective evaluation metrics of the experimental results of the proposed and compared methods on the five datasets are compared and examined. Finally, corresponding ablation studies are conducted. The experimental environments are all conducted on the Windows 11 operating system, using a computer equipped with a 2.3 GHz Core i7-12700H processor and 16GB of RAM, and the experimental platform is MATLAB 2022b.

A. Experimental Datasets

1) Los Angeles (LA) Airport Dataset: This dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer

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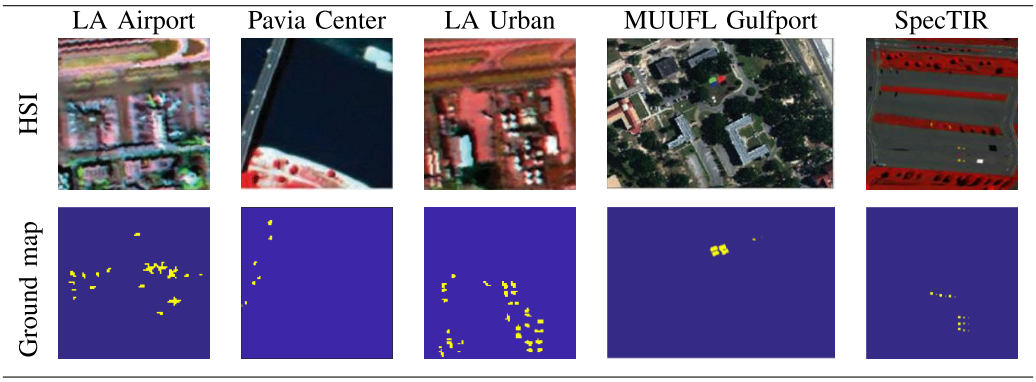


Fig. 6. False-color images and ground-truth images of the experimental datasets.

Algorithm 2 MPDA Algorithm

Input: $H \in \mathbb{R}^{M\times N\times B}$ - HSI with the size of $M \times N \times B$
Output: HSI-AD result image $F \in \mathbb{R}^{M\times N}$ - HSI-AD result image with the size of $M \times N$

1 **Initialize:** *Learning rate l, penalty factor β, number of iterations r*;
2 The guided image G is obtained by Algorithm 1;
3 Train the multilayer autoencoder based on energy-weighted skip connections to obtain reconstructed $\bar{H}$ background information according to Eq. (11);
4 Calculate the error data K between the reconstructed spectral vector $\bar{H}$ and the original spectral vector H for each pixel point;
5 Modeling of error data K using the multivariate skewed t-distribution;
6 The final AD image F is obtained by obtaining the AD score through Eq. (15);

(AVIRIS) in the LA area on November 9, 2011. The HSI is of size 100 × 100 and consists of 205 bands, with a spatial resolution of 7.1 m. It contains 170 anomalous points from airport scenes.

2) Pavia Center Dataset: This dataset was acquired in Pavia, Italy, using the Reflective Optics System Imaging Spectrometer (ROSIS-03) sensor. It comprises 102 bands with a spatial resolution of 1.3 m. For the experiment, the image of size 100 × 100 pixels was used, retaining all bands. Cars are considered anomalous targets, with 68 anomalous pixels.

3) LA Urban Dataset: This dataset captured by the AVIRIS sensor in the LA area on November 9, 2011, contains 100 × 100 pixels per spectral channel, totaling 205 bands, with a spatial resolution of 7.1 m. There are 232 anomalous points identified in residential areas.

4) MUUFL Gulfport Dataset: Acquired from the hyperspectral compact airborne spectrographic imager (CASI–1500) sensor, this dataset was captured at the University of Southern Mississippi Gulf Park Campus in November 2010. The experimental dataset in this article has a scene size of 325 × 220 and 64 spectral channels, the dataset contains 269 anomalous pixels from cloth panels.

5) SpecTIR Dataset: This dataset obtained from the SpecTIR hyperspectral airborne Rochester experiment in July 2010, consists of aerial images over Rochester, New York. The dataset size is 180 × 180 × 120 with a spatial resolution of 1 m and a spectral resolution of 5 nm. It includes 72 building anomalous pixels. All five experimental datasets used in this study are real and have been processed to remove noise and water vapor absorption bands. The reference images were manually labeled with the help of the Environment for Visualizing Images (ENVI) software. Fig. 6 shows the false-colored images of the experimental datasets along with their corresponding ground truth images.

B. Parameter Setting

In this section, three sets of parameters are involved in the MPDA algorithm, including learning rate l, penalty coefficient β, and number of iterations r. The uncertain parameters are discussed on the five datasets, with default values set to $l = 0.4$, $\beta = 1$, and $r = 300$. When analyzing one parameter, the remaining parameters are held constant. The area under the curve (AUC) values in $\text{AUC}_{(\text{PD,PF})}$ [26] is used to assess the sensitivity and contribution of different parameters to the proposed algorithm. Fig. 7 presents the line graph depicting the variation of the $\text{AUC}_{(\text{PD,PF})}$ parameter with the MPDA algorithm.

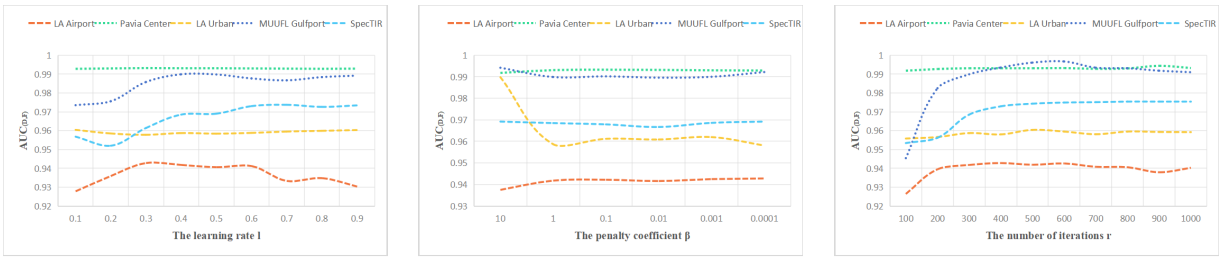
1) Learning Rate l: To determine the optimal value of l for each dataset, we set the learning rate l from 0.1 to 0.9 in increments of 0.1, while keeping other parameters β and r set to 1 and 300, respectively. Fig. 7(a) illustrates the impact of l on the detection accuracy for each dataset, with a setting of 0.3 for the first dataset and 0.4 for the rest.

2) Penalty Coefficient β: Fig. 7(b) displays a line graph of the $\text{AUC}_{(\text{PD,PF})}$ of the MPDA algorithm as a function of the parameter β. The tested coefficients are 10, 1, 0.1, 0.01, 0.001, and 0.0001. From the figure, it can be observed that we set β to 0.1 for the LA Urban dataset and 1 for the others.

3) Number of Iterations r: The number of iterations affects the quality of the reconstructed background image. The other parameters l and β are set to 0.4 and 1, respectively. The r is set within the range of 100–1000. Fig. 7(c) demonstrates the effect of the on the effectiveness of the MPDA method. According to Fig. 7(c), the number of iterations for each dataset is set to 300, 900, 300, 600, and 800, respectively.

For the uncertainty parameters in the comparison algorithms, we set fixed default values based on the original authors' recommendations. The GRX method does not have

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(a) (b) (c)

Fig. 7. Effect of different parameters on the performance of MPDA algorithm method. (a) Learning rate l. (b) Penalty coefficient β. (c) Number of iterations r.

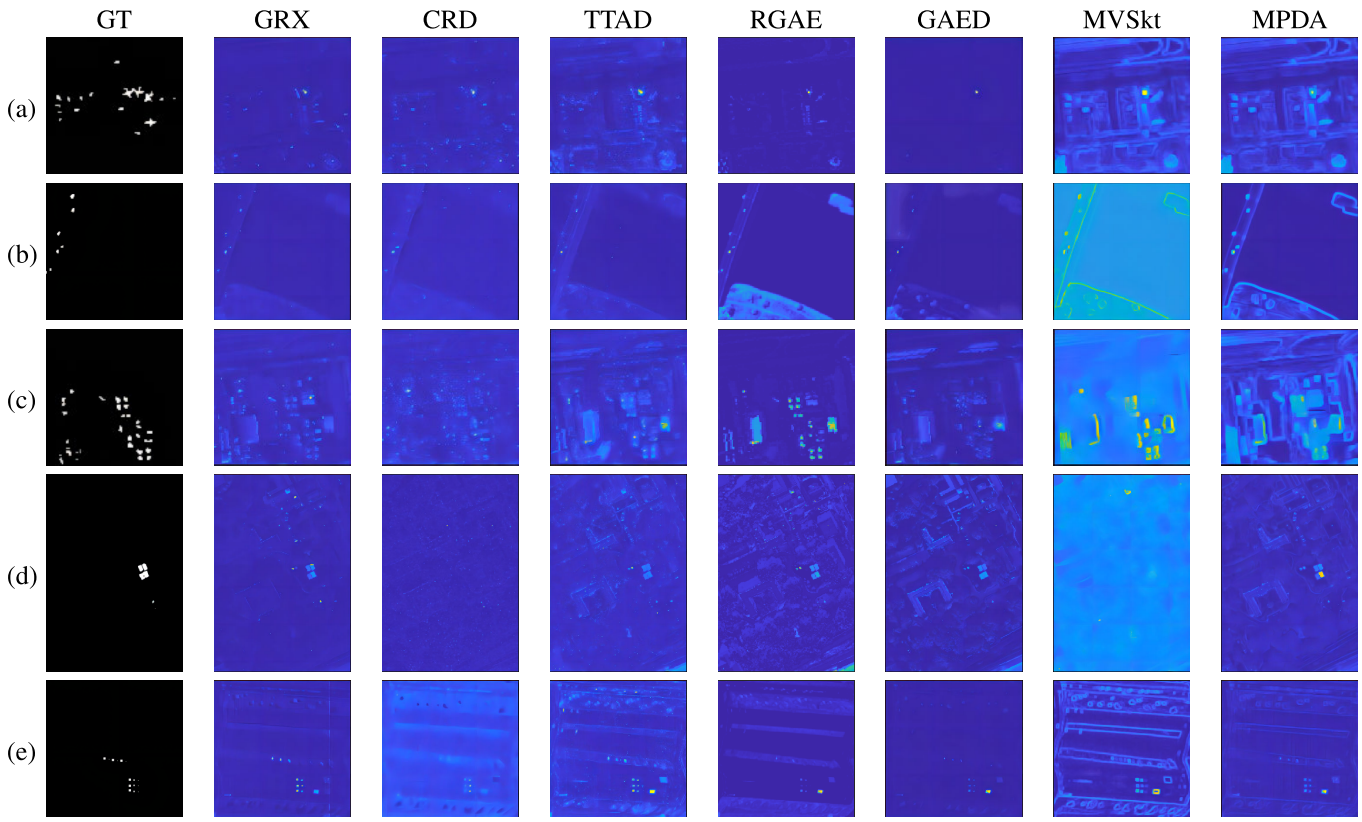


Fig. 8. 2-D colored AD maps for the five datasets. (a) LA Airport. (b) Pavia Center. (c) LA Urban. (d) MUUFL Gulfport. (e) SpecTIR.

uncertain parameters. In the collaborative representation-based detector (CRD) method, the regularization parameter λ is set to 0.01, and the size of the dual window is set to (3,5). In the tree topology-based AD (TTAD) method, the number of topological spaces $\text{Frame}_{\text{size}} = 50$ and the subsampling percentage is set to $\text{sub}_{\text{percent}} = 0.5\%$. In the RGAE method, the regularization parameter is set to 0.02, the size of the hyperpixel is set to 150, and the dimension of the intermediate hidden layer is 100. In the GAED method, the weights and biases are randomly initialized for all datasets, with a learning rate of 0.4, penalty coefficient of 1, maximum iteration of 300, and intermediate hidden layer dimension of 10. In the MVSkt method, the false positive rate FAR = 0.05.

C. Detection Performance

In this section, to evaluate the effectiveness of the proposed method, we compared it with other typical and state-of-the-art methods: GRX [8], CRD [32], TTAD [33], RGAE [22], GAED [23], and MVSkt [15]. We performed experiments on five sets of real HSIs. In addition to subjective evaluation results displayed in experimental figures, we also employed objective evaluations such as the 3-D and 2-D receiver operating characteristic (ROC) curve [26] and the target background separation boxplot [34] to visually depict detection rates and target separation effects. The $\text{AUC}_{(\text{PD,PF})}$, $\text{AUC}_{(\text{PD},\tau)}$, $\text{AUC}_{(\text{PF},\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{SNPR}, AUC_{TDBS}, and AUC_{OD} [26] are used to quantitatively assess overall detection accuracy and evaluate background suppression performance.

By comparing with the other five AD methods, we analyzed the detection performance of the MPDA method. Visual result images of different contrast algorithms on the five experimental datasets are shown in Fig. 8. The subjective evaluation results based on the color result images show that in the LA Airport dataset, the MVSkt and MPDA algorithms clearly detected anomalies, but the background contour information of both of them is more obvious, and the rest of the algorithms

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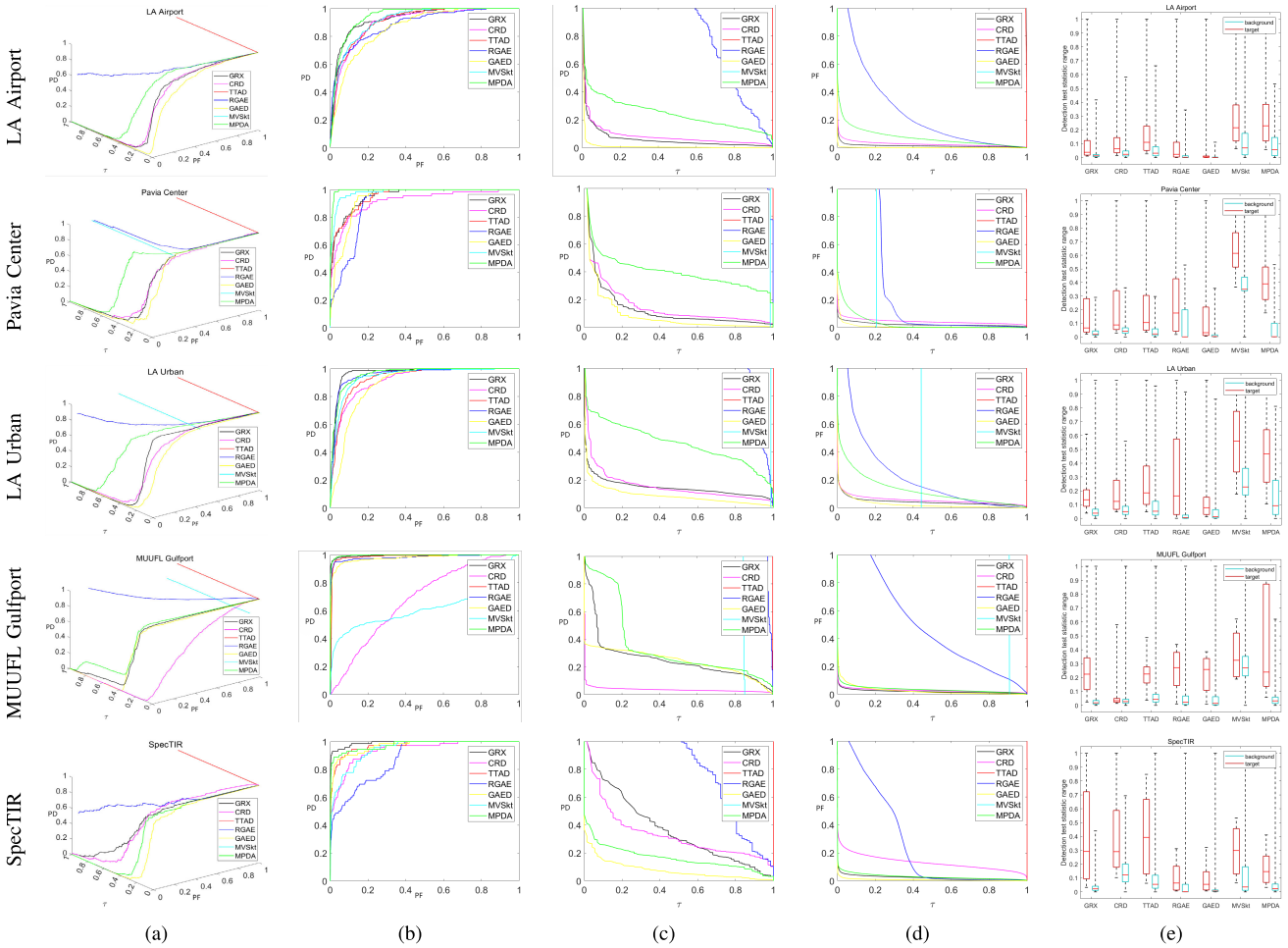


Fig. 9. Illustration of detection results. (a) 3-D ROC curves. (b) 2-D ROC (PD, PF) curves. (c) 2-D ROC (PD, τ) curves. (d) 2-D ROC (PF, τ) curves. (e) Corresponding target background separation boxplot.

detect the anomalies minimally. In the Pavia Center, the RGAE and MPDA clearly detected the anomalies, and the algorithms in this article have better detection ability, MVSkt background contour information is clearly detected, and the rest of the algorithms vaguely detected the contour information of the bridges and land. In LA Urban data, TTAD and RGAE clearly detected the anomalies, and MPDA detected the anomalies along with the fuzzy background information. In MUUFL Gulfport, the GRX and MPDA algorithms detected optimally. In SpecTIR, GRX and MPDA algorithms clearly detected the anomalies and the rest of the algorithms detected the anomalies along with the background information mixed in. In summary, the proposed method successfully detected anomalous targets and separated them from the normal targets, demonstrating ideal background suppression effects. The MPDA method obtained excellent detection results on all datasets, with final images closely resembling the reference images.

Fig. 9(a) shows the 3-D ROC curves corresponding to each detection algorithm on the experimental dataset. The algorithms in this article present high and smooth graphs under different thresholds, which demonstrates that the MPDA algorithm can effectively detect anomalous targets and presents the stability and superiority of the algorithms under a wide range of thresholds. Fig. 9(b) shows the 2-D (PD, PF) ROC curves corresponding to each detection algorithm on the experimental datasets. Curves closer to the upper left corner indicate higher accuracy in AD. Among the five datasets, CRD and GAED exhibited less ideal performance compared to other contrast algorithms. The red and light blue curves are closer to the upper left corner, indicating that GRX and MPDA achieved good detection performance, with MPDA having higher detection accuracy than GRX. Fig. 9(c) and (d) shows the 2-D ROC (PD, τ) curves and 2-D ROC (PF, τ) curves corresponding to each detection algorithm on the experimental datasets. From the figure, it can be seen that the accuracy value of the algorithm in this article is relatively better. Fig. 9(e) shows the boxplot statistics for target background separation, where anomalies are represented by red boxes, and the background by the blue box. The separability between anomalous and background targets is indicated by the gap between the red and blue boxes, with a larger gap indicating better separability. The height of the blue box represents the degree of background suppression, with a shorter blue box indicating stronger background suppression capability. From Fig. 9(e), it can be observed that in the last four experimental datasets, the gap between the red and blue boxes for the MPDA algorithm is relatively large, especially in the Pavia Center dataset, indicating that the proposed algorithm has

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TABLE II
OBJECTIVE EVALUATION OF EACH ALGORITHM FOR DATASETS

	detector	$\mathrm{AUC}_{(\mathrm{PD,PF})}$	$\mathrm{AUC}_{(\mathrm{PD},\tau)}$	$\mathrm{AUC}_{(\mathrm{PF},\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{OD}
(a)	GRX	0.9288	0.0660	**0.0145**	0.9948	**0.9144**	4.5687	0.0516	0.9804
	CRD	0.9004	0.0855	0.0281	0.9860	0.8724	3.0494	0.0575	0.9579
	TTAD	0.9095	35.905	10.4110	36.814	-9.5015	3.4488	25.4942	26.4037
	RGAE	0.8874	2.8865	0.3213	3.7739	0.5661	8.9852	2.5653	3.4527
	GAED	0.8644	0.0160	0.0015	0.8804	0.8628	**10.4040**	0.0145	0.8789
	MVSkt	0.9149	3.4e+03	1.1e+03	3.4e+03	-1.1e+03	2.9661	2.3e+03	2.3e+03
	MPDA	**0.9427**	**0.2485**	0.0728	**1.9112**	0.8699	3.4131	**0.1757**	**1.1184**
(b)	GRX	0.9538	0.1343	0.0233	1.0881	0.9305	5.7571	0.1110	1.0648
	CRD	0.9167	0.1570	0.0442	1.0737	0.8725	3.5545	0.1128	1.0295
	TTAD	0.9514	39.647	7.5605	40.599	-6.6091	5.2441	32.0873	33.0387
	RGAE	0.9042	10.0072	1.7061	10.9114	-0.8019	5.8656	8.3012	9.2053
	GAED	0.9365	0.0920	**0.0082**	1.0285	0.9283	11.2727	0.0838	1.0203
	MVSkt	0.9809	1.5e+04	9.3e+03	1.5e+04	-9.3e+03	1.6906	6.4e+03	6.4e+03
	MPDA	**0.9941**	**0.4044**	0.0288	**1.3986**	**0.9653**	**14.0266**	**0.3756**	**1.3698**
(c)	GRX	**0.9692**	0.1461	0.0437	1.1153	**0.9255**	3.3406	0.1024	1.0716
	CRD	0.9211	0.1558	0.0560	1.0769	0.8651	2.7843	0.0998	1.0209
	TTAD	0.9347	58.206	17.114	59.141	-16.179	3.4010	41.092	42.026
	RGAE	0.9569	**6.5930**	0.4665	7.5499	0.4904	**14.1340**	6.1266	7.0835
	GAED	0.8904	0.0896	**0.0261**	0.9800	0.8643	3.4257	0.0634	0.9539
	MVSkt	0.9434	3.1e+03	1.4e+03	3.1e+03	-1.4e+03	2.1980	1.7e+03	1.7e+03
	MPDA	0.9643	0.4642	0.1260	**1.1458**	0.8356	3.6837	**0.3382**	**1.2998**
(d)	GRX	0.9950	0.2544	**0.0214**	1.2494	**0.9735**	**11.8634**	0.2330	1.2279
	CRD	0.6778	0.0375	0.0291	0.7153	0.6483	1.2880	0.0084	0.6862
	TTAD	0.9859	56.533	12.741	57.519	-11.755	4.436	43.791	44.777
	RGAE	0.9802	**4.7522**	0.5873	5.7324	0.3929	8.0920	4.1650	5.1452
	GAED	0.9762	0.2394	0.0267	1.2157	0.9496	8.9799	0.2128	**1.8900**
	MVSkt	0.6168	1.6e+03	1.3e+03	1.6e+03	-1.3e+03	1.2461	326.33	326.94
	MPDA	**0.9968**	0.3520	0.0358	**1.3488**	0.9610	9.8430	**0.3163**	1.3130
(e)	GRX	**0.9914**	0.3683	0.0249	**1.3597**	**0.9665**	**14.8183**	**0.3434**	**1.3348**
	CRD	0.9373	0.3368	0.1314	1.2741	0.8059	2.5626	0.2054	1.1427
	TTAD	0.9699	105.71	17.339	106.68	-16.369	6.0970	88.378	89.348
	RGAE	0.8776	**1.4601**	0.2898	2.3377	0.5879	5.0384	1.1703	2.0480
	GAED	0.9693	0.0769	**0.0080**	1.0462	0.9613	9.6360	0.0689	1.0383
	MVSkt	0.9443	4.0e+04	9.4e+03	4.0e+04	-9.4e+03	4.2172	3.1e+04	3.1e+04
	MPDA	0.9753	0.1638	0.0305	1.1391	0.9448	5.3779	0.1334	1.1086

better separability compared to other contrast algorithms and can better separate anomalous points from the background. In the MUUFL Gulfport and SpecTIR datasets, the blue box of the MPDA is the shortest. Overall, the proposed algorithm effectively separates anomalous targets from background information.

Table II gives the AUC values for the objective evaluation metrics corresponding to the five datasets: (a) LA Airport, (b) Pavia Center, (c) LA Urban, (d) MUUFL Gulfport, and (e) SpecTIR. Table II lists the $\mathrm{AUC}_{(\mathrm{PD,PF})}$ values of different methods for the real dataset as the most important primary objective evaluation metric. By comparing the $\mathrm{AUC}_{(\mathrm{PD,PF})}$ values, it can be observed that the overall performance evaluation of MPDA is almost the best. The $\mathrm{AUC}_{(\mathrm{PD,PF})}$ values of MPDA are optimal for LA Airport, Pavia Center, and MUUFL Gulfport data. In the LA Urban and SpecTIR datasets, except for the GRX algorithm, the MPDA are all higher than the other compared algorithms.

From Table II, in the Pavia Center dataset, all the evaluation metrics of MPDA are optimal, and the overall detection metrics are the best. The LA Airport dataset is optimal for MPDA in terms of target detection probability and overall detection performance, and the metrics evaluation of the proposed algorithm is rated first among all the metrics except the $\mathrm{AUC}_{(\mathrm{PF},\tau)}$, $\mathrm{AUC}_{\mathrm{BS}}$, and $\mathrm{AUC}_{\mathrm{SNPR}}$. In the LA Urban dataset, with the three metrics of $\mathrm{AUC}_{\mathrm{TD}}$, $\mathrm{AUC}_{\mathrm{TDBS}}$, and $\mathrm{AUC}_{\mathrm{OD}}$ of the proposed in this comparison algorithms are rated first in all the metrics. In the MUUFL Gulfport and SpecTIR datasets, the AUC values of the proposed MPDA method are close to the optimal AUC values, although not all of them are optimal. The performance of the proposed MPDA method is not always the best on all the datasets in the experiment, but satisfactory results are achieved. Therefore, the proposed framework in this chapter is an effective and excellent anomaly detector.

TABLE III
RUNNING TIME OF EACH ALGORITHM FOR FIVE DATASETS

	Time(s)				
	LA Airport	Pavia Center	LA Urban	MUUFL Gulfport	SpecTIR
GRX	5.248	6.339	3.699	13.507	11.526
CRD	4.428	5.549	4.116	6.272	5.289
TTAD	7.055	11.297	7.876	31.961	14.676
RGAE	87.265	134.589	88.694	392.204	203.855
GAED	34.453	53.921	33.373	138.332	102.307
MVSkt	2.546	2.309	2.516	4.078	3.829
MPDA	59.849	110.931	61.692	308.175	177.014

The running times of the different methods in the HSI-AD experiment are shown in Table III. Among all the compared methods, GRX, CRD, TTAD, and MVSkt have the shortest computational time. These methods belong to the statistical and machine learning-based hyperspectral AD methods, which usually have low computational complexity and short training time. The RGAE, GAED, and MPDA belong to the DL-based

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TABLE IV
ABLATION STUDY OF THE PROPOSED MODELS

	$AUC_{(PD,PF)}$				
	LA Airport	Pavia Center	LA Urban	MUUFL Gulfport	SpecTIR
MAAD	0.8502	0.9059	0.8920	0.5644	0.5954
MAES	0.8716	0.9363	0.9116	0.9762	0.9692
NMST	0.8574	0.9382	0.9404	0.9776	0.9685
NGMB	0.9426	0.9928	0.9613	0.9894	0.9680
MPDA	**0.9427**	**0.9941**	**0.9643**	**0.9968**	**0.9753**

hyperspectral AD methods, which have high time complexity, and a large number of forward and reverse operations are required to train a deep network, and the higher the number of training layers, the more computational time is required, which are all relatively time-consuming. In a comprehensive subjective and objective evaluation, the MPDA algorithm proposed in this article obtains better performance but also spends an appropriate amount of running time.

D. Ablation Study

The MPDA algorithm framework consists of three branches: the guided map branch, the autoencoder branch, and the modeling of the multivariate skewed t-distribution branch, as shown in Fig. 1. To test the impact of these three parts on the AD effectiveness of the proposed algorithm, we conducted an ablation study to evaluate the performance of the proposed model. Table IV shows the $AUC_{(PD,PF)}$ for each branch, including multilayer autoencoder AD (MAAD), multilayer autoencoder with energy-weighted skip connections (MAES), MPDA without multivariate skewed t-distribution (NMST), MPDA without guided map branch (NGMB), and MPDA method on the five experimental datasets.

From Table IV, it can be seen that the $AUC_{(PD,PF)}$ of MAES is much higher than MAAD. The objective evaluation values of MAES and NMST are similar and both are lower than the method proposed in this article. NGMB is only lower than MPDA in terms of objective evaluation. This indicates that the multilayer autoencoder with energy-weighted skip connections branch, the guided map branch, and the multivariate skewed t-distribution all contribute to effective detection. The objective evaluation value of the MPDA method is superior to other branches. The experimental results verified that all three branches contribute to the proposed algorithm by providing a certain level of background suppression and better highlighting of anomalous targets, and all of them help to improve the performance of the model and the validity of the HSI.

IV. CONCLUSION

In this article, we propose an HSI-AD based on MPDA. The framework visually presents HSIs according to their frequency distribution, allowing for an easier understanding of the distribution of anomalous and background data without the need for any prior information. It effectively enhances the sensitivity to detecting anomalous targets. By coupling the autoencoder system with a multivariate probability distribution model, the framework achieves effective learning of the intrinsic features of the data without explicit data labels, discovering hidden spatial intrinsic features and enabling more effective utilization of spatial information. In addition, the architecture overcomes the inherent spectral redundancy in HSIs. Through the designed grid structure and multilayer autoencoder based on energy-weighted skip connections, the separability between anomalies and background is enhanced, further improving the accuracy of AD. Experimental validation using real HSIs captured in five different scenarios compared with typical and state-of-the-art AD algorithms confirms the strong generalization and detection capabilities of the proposed algorithm. However, MPDA currently lacks a decisive advantage in detecting anomalies in structures such as buildings. This is likely because different building structures often utilize identical or similar materials, such as concrete, glass, and metal, resulting in highly similar spectral signatures in HSIs. Consequently, HSI-AD algorithms may struggle to differentiate these subtle variations, leading to subpar performance in AD. This area requires further investigation.

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Zhenhua Mu received the M.Eng. degree from the School of Computer and Information and Technology, Liaoning Normal University, Dalian, China, in 2020, where she is currently pursuing the Ph.D. degree with the School of Geography.

Her research interests are related to anomaly detection of hyperspectral images.



Yihan Wang received the B.E. degree from the School of Computer and Information and Technology, Liaoning Normal University, Dalian, China, in 2021, where she is currently pursuing the M.Sc. degree with the School of Computer Science and Artificial Intelligence.

Her research interest includes remote sensing image processing.



Yating Zhang received the B.E. degree from the School of Computer and Information and Technology, Liaoning Normal University, Dalian, China, in 2020, where she is currently pursuing the M.Sc. degree with the School of Computer Science and Artificial Intelligence.

Her research interests include hyperspectral anomaly detection.



Chuanming Song received the Ph.D. degree from the College of Computer Science and Technology, Nanjing University, Nanjing, China, in 2010.

He is a Professor with the School of Information Engineering, Dalian University, Dalian, China. His research interests include image and video coding and multimedia digital watermarking.

Dr. Song is a member of China Computer Federation (CCF).



Xianghai Wang received the Ph.D. degree from Jilin University, Changchun, China, in 1999.

Currently, he is a Professor with the School of Computer Science and Artificial Intelligence, Liaoning Normal University, Dalian, China. His research interests include image processing, multimedia information processing, and remote sensing image processing.

Dr. Wang is a Senior Member of the China Computer Federation (CCF).

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